

used in TV sets. This laser-cut technology allows for separating CRTs quicker and cleaner than before. With this highly-efficient technology, Panasonic is poised to respond to the increasing demand for recycling of used CRT TVs, tonnes of which are expected to be thrown away as the shift to digital terrestrial broadcasting is scheduled for completion in 2011 in Japan.

As different types of glass are used in the front panel and funnel, it is essential to separate them without mixing for recycling. The conventional method uses an electrically heated wire around the joint area of the front and funnel to separate them. However, this method not only requires time for the heating process but causes thermal stress fractures by the local heating, requiring manual corrections to clean the cut surfaces.

Panasonic's CRT recycling technology utilises laser radiation to reduce the processing time with much less manual work, allowing one tube to be processed in 50 seconds—three times faster than the previous method. The laser head of the innovative system has a surface profiling function to maintain a constant distance between the focal point and surface of the glass. Coupled with the radiation energy control adjusting laser beam light intensity to the circumferential velocity, the system achieves a high-quality cut with no mixing between the front and funnel glass.

Universal phone charger

After months of negotiations with the European Commission, major producers of mobile phones have finally agreed to develop a standard for phone charging based on the Micro-USB interface. The companies that signed the MoU include Apple, LG, Motorola, NEC, Nokia, Qualcomm, Research in Motion, Samsung, Sony

Ericsson and Texas Instruments.

With this agreement, consumers will be able to charge mobile phones anywhere from the new common charger. This also means considerably less electronic waste, because people will no longer have to throw away chargers when buying new phones.

The large majority of phones will charge within two hours with the new common chargers, says the EU. Only in combination of a large battery with a small charger, the phone will charge within six hours.

Tiny, printable batteries

Researchers at the Fraunhofer Research Institution for Electronic Nano Systems ENAS have developed a battery that weighs less than 1 gram and is less



Fraunhofer tiny printable battery

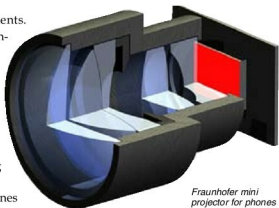
than 1 millimetre thick. The battery, which can be printed in a process similar to silk-screening shirts, could power bank cards or other slim application carriers.

"Our goal is to be able to mass produce the batteries at a price of single-digit cent range each," said Dr Andreas Willert, group manager, ENAS.

The battery is environment-friendly as it contains no mercury. Its voltage of 1.5V lies within the normal range. By placing several batteries in a row, voltages of 3V, 4.5V and 6V can also be achieved.

Mini projector for phones

Researchers at the Fraunhofer Institute for Applied Optics have developed



Fraunhofer mini projector for phones

a mini projector that promises to cut down on the flab and the large power requirements of existing phone projection technologies. The new system uses a lens system to project light and images generated by a low-power organic light-emitting diode (OLED). Use of the OLED means there is no need for a separate light source and the lens system ensures decent image reproduction.

Solar-powered, waterproof mobile phone

NTT DoCoMo has developed a solar-powered, waterproof mobile phone called Solar Hybrid (DoCoMo STYLE series SH-08A). The Solar Hybrid can be recharged with sunlight as well as electricity. Even if the battery has completely lost its charge, just 10 minutes of solar recharging via a panel on the phone's body provides approximately one minute of talk time.

The handset comes with an 8-megapixel CCD camera, and its waterproof body is resistant to water sprays from all directions and submersible to depths up to one metre.



NTT DoCoMo Solar Hybrid series phone

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